

$$= -4.73 \times 10^8 \text{ J} \quad \text{A1}$$

5 (a) (i) Constant phase difference B1

(ii) The vapour produces photons when electrons de-excite from a higher energy state to a lower energy state. B1

The de-excitation of electrons is a random process. B1

The photons emitted are produced at random time intervals and therefore the phase difference between the emitted photons is not constant. B1

(b) (i) Any 2 of the following: B2

- Waves have equal or approximately equal amplitude
- The slit separation must be of the same order of wavelength as the wave
- The distance between the two slits must be small compared to the distance between the source and screen.
- The waves from the two sources must be unpolarised or polarised in the same plane

(ii)
$$x = \frac{\lambda D}{a}$$

$$= \frac{590 \times 10^{-9} \times 2.3}{1.2 \times 10^{-3}}$$
C1

$$= 1.1308 \times 10^{-3}$$

$$= 1.1 \times 10^{-3} \text{ m (to 2sf)} \quad \text{A1}$$

(iii) Since $I \propto A^2$,

$$\frac{I_A}{I_B} = \frac{A_A^2}{A_B^2}$$

$$\frac{I}{\frac{1}{3}I} = \frac{A_A^2}{A_B^2}$$

$$A_B = \frac{1}{\sqrt{3}} A_A$$

M1

By principle of superposition, the amplitude of a dark fringe near P is

$$A = A_A - A_B$$

$$= A_A - \frac{1}{\sqrt{3}} A_A$$

$$= \left(1 - \frac{1}{\sqrt{3}}\right) A_A$$

M1

Therefore, intensity of dark fringe is:

$$\frac{I_{\text{fringe}}}{I} = \frac{A^2}{A_A^2}$$

$$I_{\text{fringe}} = \left(\frac{A^2}{A_A^2}\right) I$$

$$= \left(1 - \frac{1}{\sqrt{3}}\right)^2 I$$

C1

$$= 0.17863 I$$

$$= 0.18 I$$

A1